

Borrower Risk Scores and Loan Outcomes: A Statistical Analysis of Prosper Loan Data

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Overview

This report presents a statistical analysis of the *Prosper Loan Data* dataset, a publicly available collection of 113,937 peer-to-peer loan listings originated on the Prosper.com lending platform (Okam, 2022). The dataset is accessible at <https://www.kaggle.com/datasets/henryokam/prosper-loan-data>. The analysis examines whether a borrower's Prosper-assigned risk score, *ProsperScore*, is associated with whether a loan was ultimately completed or ended in default or charge-off. This question is worth answering because *ProsperScore* is designed to summarize a borrower's credit risk at the time a loan is originated, and testing whether it actually differentiates loans that succeed from loans that fail speaks directly to how well the platform's risk-scoring system performs in practice.

Dataset Description

The dataset represents individual loan listings on Prosper, a peer-to-peer lending marketplace that connects individual borrowers with individual lenders. Each row corresponds to a single loan and originally included 81 variables covering loan terms, borrower financial characteristics, and loan performance outcomes. After removing 1,456 records with an out-of-range *ProsperScore* value and 823 duplicate listings (identified by a repeated *ListingKey*), the cleaned dataset used for analysis contained 111,658 rows and 81 columns. The key variables examined were *LoanStatus* (the categorical outcome of interest, e.g., Completed, Defaulted, Chargedoff), *ProsperScore* (a numeric risk score ranging from 1 to 10, with 10 representing the lowest risk), and several supporting numeric variables including *BorrowerAPR*, *BorrowerRate*, *StatedMonthlyIncome*, *DebtToIncomeRatio*, and *LoanOriginalAmount*.

Methods

Before conducting any analysis, the dataset was screened for data-quality issues consistent with the initial data analysis (IDA) framework described by Lusa et al. (2024), which recommends that analysts examine data properties—including out-of-range values, duplicate records, and missingness—before proceeding to substantive statistical analysis. This screening step identified 1,456 *ProsperScore* values above the documented maximum of 10 and 823 duplicate loan listings; both were removed prior to analysis.

Descriptive statistics (means, standard deviations, quartiles, and categorical counts) were selected to characterize the central tendency, spread, and distribution of the dataset's key numeric and categorical variables before any inferential testing was performed, in keeping with the IDA recommendation that data properties be documented prior to hypothesis-driven analysis (Lusa et al., 2024).

Three visual models were chosen to represent complementary aspects of the dataset. A log-scaled bar chart was used to display the distribution of *LoanStatus* because the raw counts across categories span several orders of magnitude, which a linear scale would obscure. A faceted histogram was used to compare the distribution of *ProsperScore* across loan outcomes because histograms are well suited to comparing the shape of a numeric distribution across groups. A correlation heatmap was used to examine relationships among the dataset's numeric predictors, since it concisely summarizes many pairwise relationships at once and can reveal redundancy among variables.

An independent-samples *t*-test was selected to test for a difference in mean *ProsperScore* between completed loans and failed loans (defaulted or charged off) because the analytical question compares a numeric variable across two independent groups. Specifically, Welch's *t*-test was used rather than Student's *t*-test. Delacre et al. (2017) argue that Welch's *t*-test should be

used by default when comparing two independent groups, because it maintains accurate Type I error control under unequal variances and performs comparably to Student's test when variances are equal, whereas Student's test can be severely biased when the equal-variance assumption is violated. This assumption was checked directly with Levene's test, which indicated that the two groups' variances were significantly unequal ($p < .0001$), confirming that Welch's correction was appropriate for this dataset. The t -test assumes independent observations, which is reasonable here since each row represents a distinct loan listing, and approximate normality of the sampling distribution of the mean, which is well supported given the large sample size in each group ($n = 19,621$ and $n = 6,340$, respectively).

Results

As shown in Figure 1, *LoanStatus* is heavily imbalanced: most loans are either *Current* or *Completed*, with *Defaulted* and *Chargedoff* loans making up a smaller but still substantial share of the dataset. Descriptive statistics showed that *ProsperScore* had a mean of 5.87 ($SD = 2.30$) on its 1–10 scale, indicating that most borrowers were assigned a moderate risk rating rather than an extreme one.

Figure 2 compares the distribution of *ProsperScore* across the three loan outcomes of interest. Completed loans show a right-skewed distribution that peaks near the safer end of the scale (around 8), while defaulted and charged-off loans show a flatter distribution that peaks lower, around 5–6, indicating a broader spread into riskier score values.

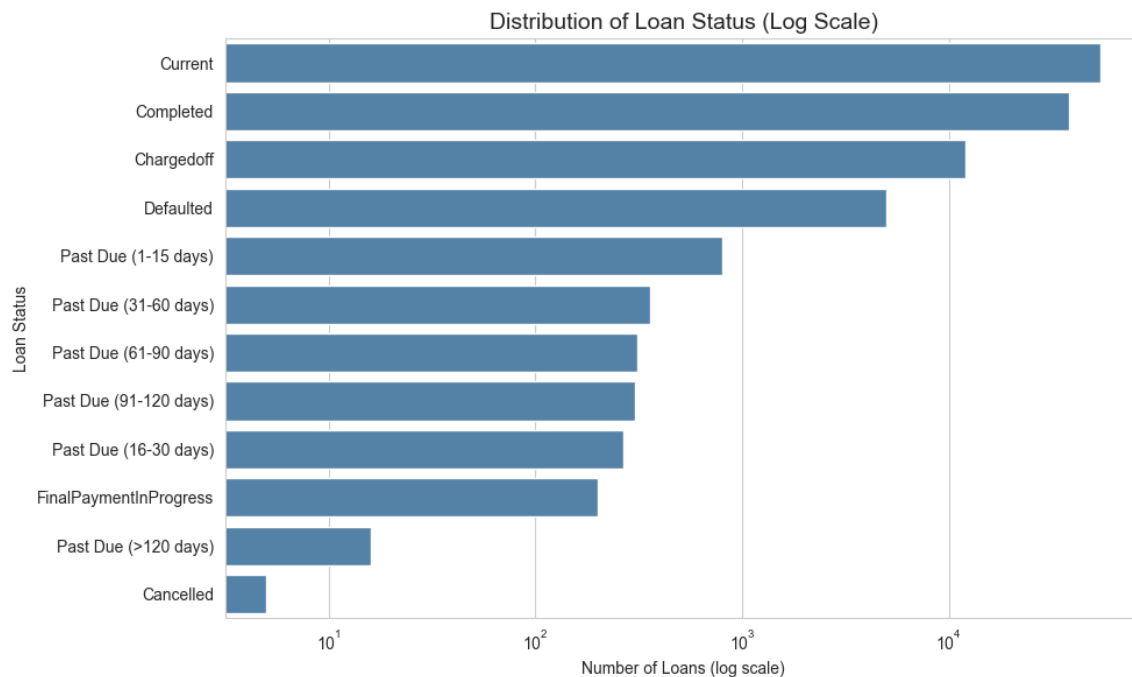
Figure 3 shows the correlation heatmap of key numeric predictors. *ProsperScore* was strongly negatively correlated with both *BorrowerAPR* ($r = -0.66$) and *BorrowerRate* ($r = -0.64$), consistent with a risk-based pricing model in which safer borrowers are charged lower interest rates. An unexpected finding was that *BorrowerAPR* and *BorrowerRate* were correlated

at $r = 0.99$ with each other, indicating extreme multicollinearity between these two variables; because APR is derived from the base rate plus fees, the two carry largely redundant information and should not both be treated as independent predictors in any model built from this dataset.

The hypothesis test compared mean *ProsperScore* between completed loans ($n = 19,621$, $M = 6.53$, $SD = 2.30$) and failed loans ($n = 6,340$, $M = 5.43$, $SD = 2.16$). Welch's t -test found this difference to be statistically significant, $t(11,328.79) = 34.80$, $p < .001$, leading to rejection of the null hypothesis. The effect size was calculated as Cohen's $d = 0.49$, which by conventional thresholds ($0.2 = \text{small}$, $0.5 = \text{medium}$, $0.8 = \text{large}$) represents a medium effect, indicating that the difference is not only statistically detectable but also practically meaningful.

Figure 1

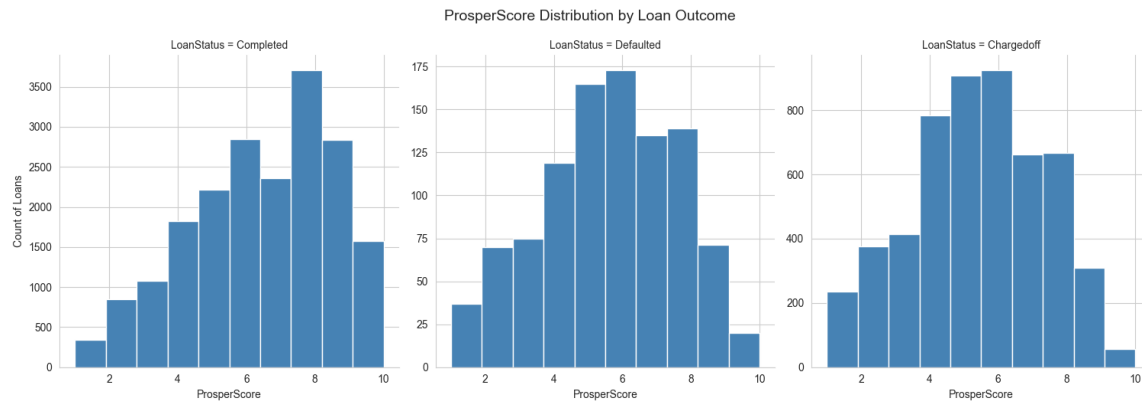
Distribution of Loan Status (Log Scale)



Note. Counts are shown on a logarithmic x-axis because loan status categories span several orders of magnitude.

Figure 2

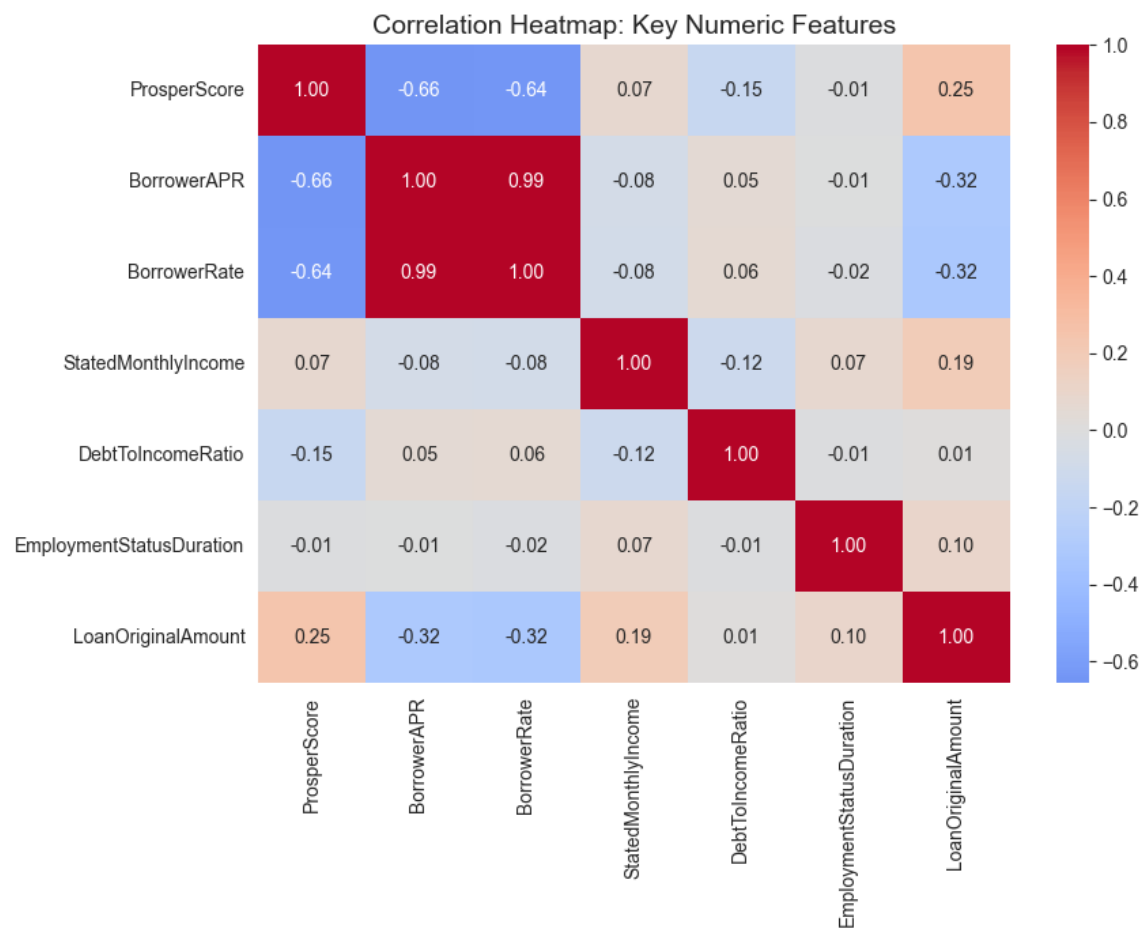
ProsperScore Distribution by Loan Outcome



Note. Panels show Completed, Defaulted, and Chargedoff loans, from left to right.

Figure 3

Correlation Heatmap of Key Numeric Features



Note. Values are Pearson correlation coefficients.

Interpretation for a Non-Technical Audience

In plain terms, Prosper’s risk score does meaningfully distinguish borrowers who go on to repay their loans from those who default. Borrowers whose loans were completed had an average risk score of about 6.5 out of 10, compared to about 5.4 out of 10 for borrowers whose loans defaulted or were charged off. That roughly one-point gap might sound small, but with over 25,000 loans in the comparison, we can be very confident it is not due to chance, and the size of the gap is large enough to matter in practice, not just on paper. In other words, Prosper’s scoring system is doing meaningful work: borrowers it rates as safer really do default less often. At the same time, the overlap between the two groups’ score distributions (visible in Figure 2) shows that the score is far from a perfect predictor — many borrowers with mid-range or even fairly high scores still defaulted, so the score should be understood as one useful signal among several, not a guarantee of repayment.

Limitations and Potential Bias

One notable dataset limitation is the amount of missing data in *ProsperScore*: 29,084 of 111,658 loans (about 26%) had no score recorded, because Prosper only began assigning this score to loans originated after July 2009. This missingness is not random with respect to time, which the IDA framework emphasizes as an important property to document, since a data-screening step that overlooks such structural missingness risks silently reducing the analysis to loans from a narrower and more recent time period than the full loan history the dataset appears to represent (Lusa et al., 2024). Because the hypothesis test in this analysis excluded rows with a missing *ProsperScore*, its conclusions technically apply only to loans originated after mid-2009 rather than to the full 2005–2014 span of the raw dataset.

A second, more concerning source of potential bias is that the loan outcomes and pricing captured in this dataset may themselves reflect discriminatory lending patterns rather than purely risk-based decisions. Pope and Sydnor (2011) found direct evidence of racial discrimination on the Prosper platform: loan listings that included a picture of a Black borrower were substantially less likely to be funded than observably similar listings from White borrowers, and Black borrowers who did receive funding paid meaningfully higher interest rates than comparable White borrowers. Because this dataset does not include the borrower demographic information used in that study, it is not possible to test directly for similar bias here, but the finding raises the possibility that some of the variation in *LoanStatus* and *BorrowerAPR* observed in this analysis reflects lender or platform-level bias rather than borrower risk alone, and that *ProsperScore* itself, if built in part from historical outcomes shaped by such bias, could partially encode it. This possibility should be kept in mind when interpreting the strength of the relationship reported here.

References

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